BKEMS Prototype Store: ESP32 Pin Tables (draft rev 3, Oct 5 2026)

This goes with `BKEMS-Wiring-Diagram.pdf` (rev 3) and `BKEMS-Wiring-Directions.md`.

Rev 3 architecture: one **store node** ESP32 inside the building, plus **one ESP32 at each rooftop unit** (RTU node, up to 5). Each RTU node reads its own thermostat calls locally. Power for each RTU node comes from that unit's 24VAC transformer through a ~**1A inline fuse** into a buck converter. **C is required** (buck return + optocoupler reference) and is taken at the unit – never needed at the thermostat.

- **From Andy's notes:** store-node relay pins (GPIO25/26/27/33), relay wire colors, and the MQTT contract v1.0 relay keys.
- **Proposed:** suggested pins. They aren't final and can change during the bench test.

1. Store node: ESP32 DevKit (Freenove breakout), in the building

ESP32 pin	Connects to	Wire color	Contract key / purpose	Notes (proposed vs from Andy's notes)
5V / VIN	5V bench supply (+)	Red	Board power	Bench supply or USB. The same 5V rail feeds the 4-ch relay VCC.
3V3	DS18B20 red leads; 4.7k pull-up; photocell module VCC; door optocoupler output side (if needed)	Red	Sensor power	3.3V only. Never put 5V or 24VAC here.
GND	5V supply (-); relay board GND; DS18B20 black leads; sensor GNDs; optocoupler output GND	Black	Common ground	**Required:** relay GND must be tied to ESP32 GND.
GPIO25	Relay IN1	Purple	`relay_parking_lot`	From Andy's notes. Load on the NC contact (fail-safe: lights stay on).
GPIO26	Relay IN2	Orange	`relay_exterior_signs`	From Andy's notes. Load on the NC contact (fail-safe).
GPIO27	Relay IN3	Blue	`relay_hvac_shed_stage1`	From Andy's notes, kept for now. **Rev 2+:** per-RTU stage-1 shed may move to each RTU node's own Y1 shed relay. If it does, IN3 becomes a reserve or store-level channel.
GPIO33	Relay IN4	White	`relay_spare_ch4`	From Andy's notes. Spare channel on the NC contact.
GPIO4	DS18B20 one-wire bus (yellow data leads), with 4.7k pull-up to 3V3	Yellow	Walk-in cooler, freezer, dining room, duct supply air temps	**Proposed.** All 4 probes share one bus; identify each by ROM address.
GPIO34	Photocell / light sensor analog out	Green (suggested)	Outdoor light level (lot/sign logic)	**Proposed.** Input-only ADC1 pin, so no conflict with Wi-Fi.

ESP32 pin	Connects to	Wire color	Contract key / purpose	Notes (proposed vs from Andy's notes)
GPIO32	AD2000-M door panel status/alarm, through an optocoupler	Gray (suggested)	Back door status / alarm	**Proposed.** The panel keeps the locking. The ESP32 only monitors it. Use an internal pull-up.
Wi-Fi (no pin)	Raspberry Pi gateway (Mosquitto + edge agent)	n/a	MQTT `bkems/{deviceKey}/telemetry` (up, ~30 s) and `bkems/{deviceKey}/cmd` (down); deviceKey = `store-{id}-gw`	From MQTT contract v1.0. The Pi forwards to the Command Center (Mac mini now, bkems.info later).

Store relay board (Inland 4-ch, 5V, SRD-05VDC)

Relay pin	Connects to	Wire color	Notes
VCC	5V bench supply	Red	Keep the JD-VCC jumper on for bench use.
GND	ESP32 GND (common)	Black	Required.
IN1-IN4	GPIO25 / 26 / 27 / 33	Purple / Orange / Blue / White	From Andy's notes. Board is active-LOW: drive LOW to energize the coil, which opens NC and sheds the load.
NC / COM	Loads	n/a	NC is fail-safe: if the ESP32 or server is down, the coil is off and the load keeps running.

12V supply: needed only if the SainSmart 16-ch or the red 8-ch (12V relay) boards are used instead of the 4-ch board.

2. RTU node: ESP32 at each rooftop unit (x up to 5), all proposed

Power comes from a 24VAC-to-5VDC buck converter fed **straight from that RTU's own 24VAC transformer** through a ~**1A inline fuse** (tap after any existing R-side fuse/breaker). The original thermostat stays primary. The RTU node only watches the calls and can drop cooling stages through NC shed relays.

C is required. Land C on the unit's own terminal board / transformer common. Use that same C for the buck return and every optocoupler AC-side return. You never need a C wire to the thermostat.

ESP32 pin	Connects to	Wire color	Purpose	Notes
5V / VIN	Buck converter 5V out (at the RTU)	Red	Node power	**Proposed.** Buck input = RTU transformer R / C (24VAC) via ~1A fuse. The same 5V feeds the shed relay coils.
3V3	DS18B20 red; 4.7k pull-up; optocoupler output pull-ups (if external)	Red	Logic power	**Proposed.**
GND	Buck 5V (-); relay GND; optocoupler output GND; DS18B20 black	Black	Common ground	**Proposed.** Common GND is required.
GPIO32	24VAC optocoupler #1 output (thermostat **W**)	Suggested: white	W heat call	**Proposed.** INPUT_PULLUP; LOW = call active. The AC opto pulses at 60 Hz, so filter/debounce in firmware (~50-100 ms).
GPIO33	24VAC optocoupler #2 output (thermostat **Y1**)	Suggested: yellow	Y1 cool stage 1	**Proposed.** INPUT_PULLUP; same filtering.
GPIO27	24VAC optocoupler #3 output (thermostat **Y2**)	Suggested: yellow/white	Y2 cool stage 2	**Proposed, optional:** only if the unit has 2-stage cool. INPUT_PULLUP.
GPIO25	24VAC optocoupler #4 output (thermostat **G**)	Suggested: green	G fan call	**Proposed.** INPUT_PULLUP.
GPIO13	24VAC optocoupler #5 output (thermostat **W2**)	Suggested: brown	W2 heat stage 2	**Proposed, optional:** only if W2 is available. INPUT_PULLUP.
GPIO26	Stage-1 shed relay IN (5V relay, NC contact in series with **Y1**)	Suggested: blue	Y1 shed	**Proposed.** Coil off = NC closed = Y1 passes normally (fail-safe). Interrupts the Y1 cooling call only.
GPIO19	Stage-2 shed relay IN (5V relay, NC contact in series with **Y2**)	Suggested: orange	Y2 shed	**Proposed, optional:** 2-stage units only. Coil off = NC closed = Y2 passes normally (fail-safe).
GPIO4	DS18B20 supply / return air probe(s), with 4.7k pull-up to 3V3	Yellow (probe lead)	Supply / return air temp	**Proposed, optional.**

ESP32 pin	Connects to	Wire color	Purpose	Notes
Wi-Fi (no pin)	Raspberry Pi gateway	n/a	MQTT telemetry/cmd for this RTU	**Proposed.** Suggested device key: `store-{id}-rtu{n}` (e.g. `store-791-rtu1`). Awaiting Andy's OK before firmware.

Cooling staging and shed rules (RTU node)

- **Staging is always Y1 first.** Y2 comes on only if Y1 can't carry the cooling load. **Y2 is never energized without Y1.**
- **Shed order:** drop **Y2 first**, then Y1.
- **Restore order:** reverse, so **Y1 first**, then Y2.
- **300 s anti-short-cycle applies per stage.** Once a stage drops, it can't restart for at least 300 s, and each stage has its own timer. If the RTU node sees Y2 called without Y1, flag it as a fault, not a valid state.
- Both shed relays are wired NC, in series with their own Y wire. If the RTU node loses power or crashes, both Y calls pass straight through and the thermostat runs the unit as it does today.

Call labels (quick reference)

Terminal	Meaning
R	24VAC hot
C	24VAC common (required at the unit for BKEMS)
W	Heat
G	Fan
Y1	1st stage cooling
Y2	2nd stage cooling (optional; only after Y1)
W2	2nd stage heat (optional)

3. Room sensor plate: ESP32-C3 (x4, one per RTU), all proposed

ESP32-C3 pin	Connects to	Wire color	Purpose	Notes
5V in	Screw terminal "5V", from the 24VAC-to-5VDC buck located AT the RTU	Red	Plate power	**Proposed.** The run carries 5VDC, not 24VAC (CAT5 if needed). Check that the voltage at the plate stays above ~4.5V during Wi-Fi bursts. Battery is not practical at a 30 s publish interval.
GND	Screw terminal "GND" (buck -); DS18B20 black; LED cathodes	Black	Common ground	**Proposed.**

ESP32-C3 pin	Connects to	Wire color	Purpose	Notes
3V3	DS18B20 red; 4.7k pull-up	Red	Sensor power	**Proposed.**
GPIO4	DS18B20 data, with 4.7k pull-up to 3V3	Yellow	Room temperature	**Proposed.**
GPIO5	Green LED (fan), with series resistor (~330 ohm)	Green	Fan status	**Proposed.**
GPIO6	Red LED (heat), with series resistor (~330 ohm)	Red	Heat status	**Proposed.**
GPIO7	Blue LED (cool), with series resistor (~330 ohm)	Blue	Cool status	**Proposed.**
Wi-Fi (no pin)	Raspberry Pi gateway	n/a	MQTT room temperature to the matching RTU logic	**Proposed.** No buttons on the plate. Optional small display later.

Safety

- Never put 24VAC on an ESP32 pin. 24VAC goes only to optocoupler inputs and buck converter inputs.
- Use a common GND between each relay board and its ESP32.
- All relays are wired normally closed (NC), so loads and Y calls keep running if an ESP32 or the server is down.
- Put a ~1A inline fuse on our feed from the RTU transformer so a short in our wiring blows our fuse instead of the unit's control fuse.
- This is a bench-only build. No compressor control. The original thermostat stays primary.